

APPENDIX B: Expert Validation Interview Transcript

This section shows the interview questions and the art experts' responses after being presented the predicted results in Appendix A.

The researchers interviewed art conservators based in Sweden.

Date: February 12, 2025

Question 1: Do you agree with the classification of the correctly identified images?

Interviewee 1: Yeah, I'd say you did a good job! But some of them are kind of a mix between styles. Art doesn't always fit into boxes. That's art, though, you shouldn't make it black and white when so much is grey area.

Interviewee 2: Yes, the classifications align with the respective movements for the most part. Here Northern Renaissance works are quite detailed. The Pop Art works reflect the bold, graphic styles associated with that particular movement. They are all correctly identified.

Interviewee 3: Yeah, for the most part! The Cubism pieces I'd say are right. Some of them can be seen as a combination though, so it's pretty hard to say with 100% certainty.

Question 2: Are the characteristics of the artworks in the correctly classified set consistent with your knowledge of those styles?

Interviewee 1: Yes, but like I also just mentioned, some pieces don't just belong to one style. I'd say most of them are clear in this case, but others feel more fluid. Artists experiment and cross boundaries, and you can't just define put art in a single movement box. One piece can be more than one style, both can be true.

Interviewee 2: Absolutely. As previously mentioned, the correctly classified pieces exhibit precisely the attributed one would expect. Let's Realism works for example. They

authentically reflect everyday subjects. Or the Ukiyo-e pieces here, show the lines typical of their woodblock prints. The Rococo pieces maintain their light, decorative qualities, emphasizing elegance and ornate detail. The characteristics are well reflected.

Interviewee 3: Definitely. The High Renaissance works show that perfect balance and harmony you'd expect, and the Minimalist pieces are super clean and stripped down, like textbook examples. But even in the correctly classified ones, art doesn't always fit neatly into one box, you know?

Question 3: How well do the model's predictions align with historical and stylistic contexts for these images?

Interviewee 1: The model's predictions are pretty solid, but again, some works could easily fit into more than one category. Take abstract expressionism and color field painting, for example. There was this one landscape that had the expressive brushstrokes of abstract expressionism but also leaned heavily on broad, solid fields of color. Even the pop art piece? You could argue it shares elements with abstract art due to its bold, simplified forms. Art often moves across style lines, and the model could reflect that flexibility more.

Interviewee 2: While the model does well in clear-cut cases, it struggles with styles that share superficial visual traits. For example, it often confuses Early Renaissance with High Renaissance, which makes sense given their overlapping techniques, but it ignores the more symbolic, structured aspects that distinguish the latter form of Renaissance which is High Renaissance.

Interviewee 3: Overall the classification is good with very distinct styles. But some similar styles are mislabeled. Like with Early Renaissance and Northern Renaissance,

both are so focused on detail and religious themes. The regional differences matter. The model just misses cultural context or the artist's personal history. Humans can see those nuances, but the model can't, even though it would help in making better predictions.

Question 4: For the misclassified images, what key differences do you see between the model's classification and your own understanding of the artwork's style?

Interviewee 1: Honestly, I'd say these classifications make sense. Some pieces look a lot like other styles. So, I get why the model tripped up. But if you know what the works show, and when it was made it's much easier to classify them. The historic context makes a huge difference.

Interviewee 2: Expressionist pieces recognized as Cubism actually have strong emotional intensity. They are more free and have a fluid style. I think Art Nouveau was misclassified with a lot of variety. Renaissance, symbolism, these styles have more organic, flowing lines. The model misinterpreted Pointilism for Impressionist pieces. I believe that your model is unable to distinguish the dot technique and small brushstrokes.

Interviewee 3: A lot of the misclassifications might come from surface similarities. Minimalism, Abstract Expressionism, and Cubism got misinterpreted here and there. Minimalism is about simplicity. Abstract Expressionism is quite chaotic and expressive, so polar opposites. Cubism uses different shapes and forms.

Question 5: Can you identify specific features like brushstrokes, color palette, historical context) that may have led to these misclassifications?

Interviewee 1: Definitely. Features like brushstrokes, color choices, and even compositional techniques can make a huge difference in classification. But apart from

just visuals, understanding intent and the cultural background paints a clearer picture of the style, no pun intended (laughs).

Interviewee 2: I'd say your model seems to focus on surface-level features, such as color application and line work. It doesn't see the compositional and thematic aspects of the art pieces, which a human would be able to see, but a model is not able to see. Expressionist works may look like Cubism in some cases. But truly, they are driven by emotional expression rather than intellectual deconstruction. These are things that, I assume, your model does not take into account.

Interviewee 3: For sure. The model seems to latch onto visual stuff like bold colors or geometric shapes, without thinking about why they are there. For example, Fauvism and Expressionism use colors both intensely. Fauvism is supposed to be much more joyful though. Expressionism goes deeply emotional and sometimes even darker themes.

Question 6: Do the misclassified images belong to closely related styles? If so, what characteristics do you think the model missed when distinguishing these styles?

Interviewee 1: Yeah, for sure—lots of the misclassified pieces are from styles that are super close to each other. I see the model struggles when styles overlap. A few are way off, though. Then the model is just missing context, like the artist, when they lived, and signature techniques.

Interviewee 2: Yeah, many misclassifications involve closely related movements. Early Renaissance and High Renaissance have some obvious similarity, as they are historically intertwined. It's the same for Cubism and Expressionism, and Rococo and Baroque. They have visual similarities. Your machine learning model doesn't really consider themes or

historic context. These pieces of context are crucial when it comes to differentiating these styles.

Interviewee 3: Yeah, High Renaissance and Northern Renaissance look similar, culturally they are different culturally. Impressionism and Post-Impressionism also got mixed up right? That makes sense, because they are similar, but Post-Impressionism has more structure and meaning behind it. The model seems to miss out on the historical progression of these styles.

Question 7: Were there any instances where the artwork presented overlapping elements from multiple styles, and the model struggled to assign the correct labels?

Interviewee 1: Oh yeah, definitely. There were quite a few pieces that had overlapping elements from multiple styles, and you could tell the model didn't know what to do with that. That's fair, though. Some works are supposed to be a bit ambiguous. Maybe the model could improve by ranking multiple styles by similarity, instead of just a single label.

Interviewee 2: Definitely. Some of the Post-Impressionist pieces show the loose brushwork of Impressionism but include symbolic content. I can see why this would confuse the model. Or take a look at Pop Art here. Works from the Abstract Expressionism movement or Color Field Painting are known to take inspiration from other styles and combine them.

Interviewee 3: Oh, for sure. Some Cubist works have aspects that they have taken from Expressionism. You see, it has strong emotional contents and less rigid geometry, like this here (points to the paper). It's tricky because a lot of modern and contemporary art doesn't stick to just one style. They're all borrowing from each other.

Question 8: For each misclassified image, how confident are you that the model has made a clear mistake, or is there ambiguity in the classification that could be addressed with more refined input?

Interviewee 1: It really depends on the piece. In certain cases the model's categorization makes sense, even if it's not perfect. Other times, it's really way off like it's missing some key context. In a few cases, I could see where it was coming from, but there were definitely some things where it felt like a total misfire.

Interviewee 2: Some mistakes are understandable, such as the confusion between Impressionism and Post-Impressionism. However, I am at a loss for some classifications. Expressionism as Art Nouveau, that's a clear error. If you can include more historical pieces and maybe refine the model's understanding of thematic content, that could improve its performance, although I have no idea how to put that into a model. But that is your job (laughs).

Interviewee 3: Some mistakes are obvious. Expressionism as Impressionism? That's just wrong. Others make sense. Styles blend easily, like Fauvism and Expressionism, I see why the model got confused. I think that if the model would consider historical context and artist intent, it would be better, but yeah, art is messy, so sometimes there's no perfect answer.